TASK 4: RESPONSIBLE AI AND MODEL INTERPRETATION

1. Objective

The objective of this project is to interpret machine-learning predictions, examine potential bias and fairness risks, and recommend responsible AI practices using feature importance and SHAP explanations.

2. Tools and Technologies

- Python
- Google Colab
- Pandas and NumPy
- Matplotlib and Seaborn
- Scikit-learn
- SHAP

3. Dataset Description

The Breast Cancer Wisconsin dataset provided by Scikit-learn was used. It contains 569 records and 30 numerical features. The target variable has two classes: malignant and benign.

4. Methodology

The dataset was divided into training and testing sets using an 80:20 split. A Random Forest classifier with 200 trees was trained on the training data. The model was evaluated on the test data using accuracy, precision, recall, and F1-score.

5. Model Interpretation

Feature importance was calculated using the trained Random Forest model. A bar chart was generated to identify the ten features with the highest importance scores.

SHAP was also used to explain model predictions. These explanations help show how feature values contribute to the model's output. Feature importance and SHAP describe model behavior but do not establish causal relationships.

6. Model Evaluation

The model was evaluated using test accuracy and a classification report. The report includes precision, recall, F1-score, and support for each target class.

Actual test accuracy: [Enter the value from Google Colab]

Insert the classification report or a screenshot of the evaluation output here.

7. Bias and Fairness Assessment

Class-wise precision, recall, and F1-score were compared to examine differences in model performance between the two target classes.

The dataset does not include the demographic attributes needed for a full demographic fairness audit. Therefore, this project does not establish whether the model is fair across demographic groups.

8. Responsible AI Recommendations

1. Evaluate the model on representative and appropriate datasets.
2. Investigate errors separately for each class.
3. Use suitable demographic data and fairness metrics when legally and ethically justified.
4. Review feature explanations with qualified domain experts.
5. Monitor model performance when data or conditions change.
6. Do not use this educational model for medical diagnosis or treatment decisions.

9. Conclusion

This project demonstrates feature importance, SHAP-based model interpretation, class-wise performance evaluation, and responsible AI risk assessment. It highlights the importance of transparency, careful evaluation, and acknowledging the limitations of a model and its dataset.

10. GitHub Repository

https://github.com/PraneethReddy151106/InternSpark-Task4-Responsible-AI

11. Evidence

Insert screenshots from Google Colab showing:

1. Model evaluation and classification report.
2. Feature importance table and chart.
3. SHAP explanation plot.
4. Class-wise performance table and chart.
5. Responsible AI assessment table.

Top 10 Most Important Features:

	Feature	Importance
22	worst perimeter	0.133100
23	worst area	0.128052
27	worst concave points	0.108107
7	mean concave points	0.094414
20	worst radius	0.090639
0	mean radius	0.058662
2	mean perimeter	0.055242
3	mean area	0.049938
6	mean concavity	0.046207
26	worst concavity	0.035357

```
Model training completed!
Test Accuracy: 95.61%

Classification Report:
              precision    recall  f1-score   support

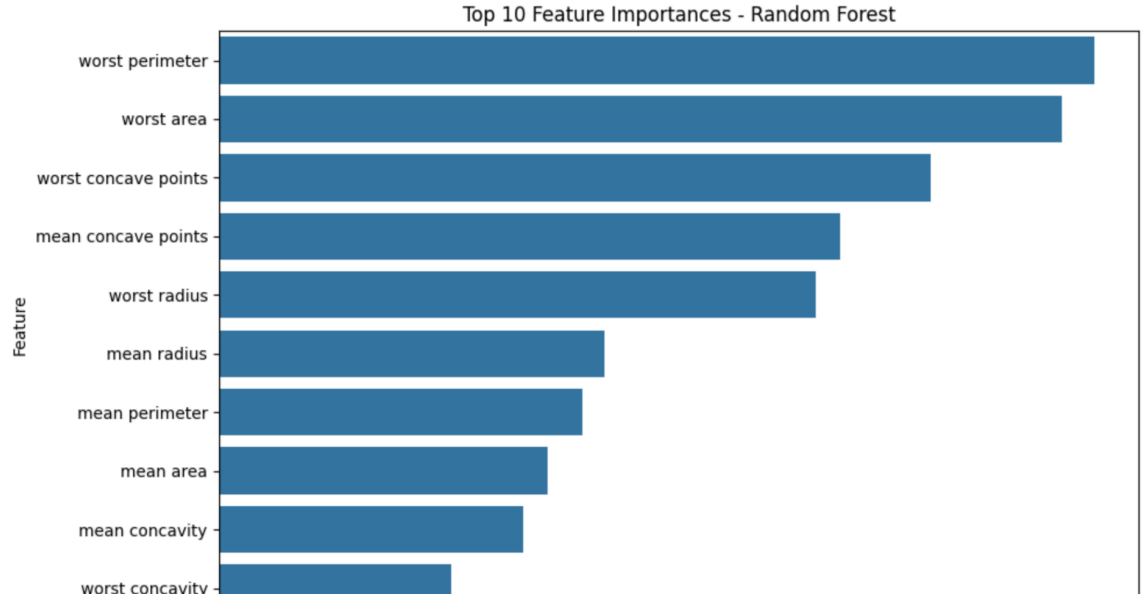
   malignant       0.95      0.93      0.94        42
      benign       0.96      0.97      0.97        72

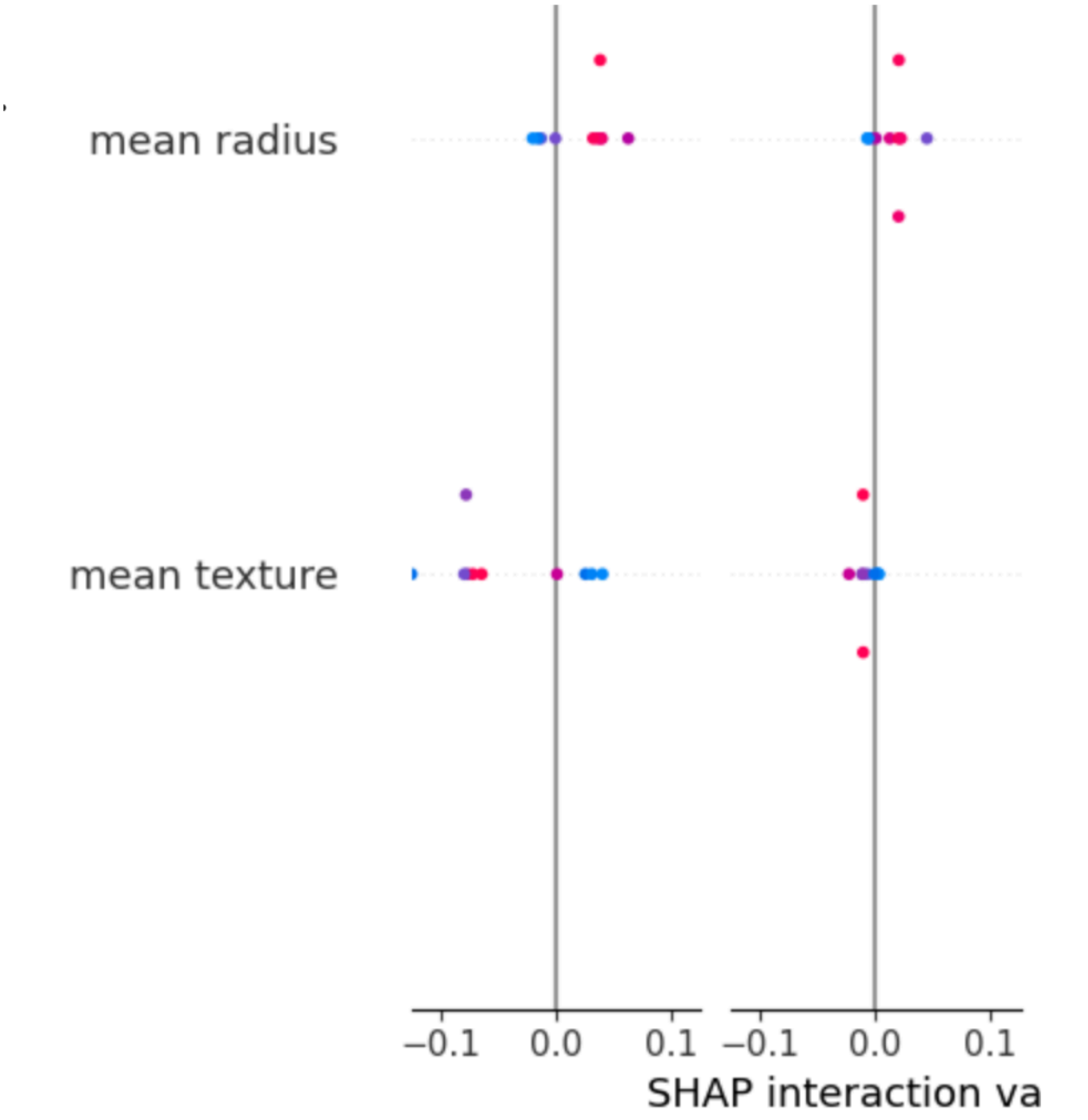
    accuracy                           0.96       114
   macro avg       0.96      0.95      0.95       114
weighted avg       0.96      0.96      0.96       114
```

```
Dataset loaded successfully!
Number of rows: 569
Number of features: 30
Target classes: [np.str_('malignant'), np.str_('benign')]
```

	mean radius	mean texture	mean perimeter	mean area	mean smoothness	mean compactness	mean concavity	mean concave points	mean symmetry	mean fractal dimension	...	worst radius	worst texture	worst perimeter	worst area
0	17.99	10.38	122.80	1001.0	0.11840	0.27760	0.3001	0.14710	0.2419	0.07871	...	25.38	17.33	184.60	2019.0
1	20.57	17.77	132.90	1326.0	0.08474	0.07864	0.0869	0.07017	0.1812	0.05667	...	24.99	23.41	158.80	1956.0
2	19.69	21.25	130.00	1203.0	0.10960	0.15990	0.1974	0.12790	0.2069	0.05999	...	23.57	25.53	152.50	1709.0
3	11.42	20.38	77.58	386.1	0.14250	0.28390	0.2414	0.10520	0.2597	0.09744	...	14.91	26.50	98.87	567.7
4	20.29	14.34	135.10	1297.0	0.10030	0.13280	0.1980	0.10430	0.1809	0.05883	...	22.54	16.67	152.20	1575.0

5 rows × 30 columns




mean radius
mean texture
−0.1
0.0
0.1
−0.1
0.0
0.1
SHAP interaction va

Class-wise model performance:

	Class	Precision	Recall	F1-score	Support
0	malignant	0.9512	0.9286	0.9398	42
1	benign	0.9589	0.9722	0.9655	72



	Risk	Finding	Mitigation
0	Demographic bias	Demographic attributes are not available in th...	Evaluate representative data with appropriate ...
1	Unequal class performance	Compare class-wise precision, recall and F1-sc...	Investigate errors for each class and validate...
2	Feature interpretation	Feature importance and SHAP explain model beha...	Review explanations with domain experts and te...
3	High-stakes use	The model is trained on an educational dataset...	Do not use this model for medical diagnosis or...

Next steps: Generate code with `fairness_assessment` New interactive sheet